SocialMind AI: User Manual

Advanced Semantic Search & Intelligent Filtering Platform

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Project Type: AI-Powered Social Media Intelligence
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Internship Domain: Artificial Intelligence & Machine Learning

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1. Introduction

SocialMind AI is a professional-grade, AI-powered web application designed to overcome the limitations of traditional keyword-based social media searching. By transitioning from lexical matching to semantic intent analysis, the platform allows users to query content based on meaning and context across Instagram, Facebook, and LinkedIn.

Project Overview

The system leverages high-dimensional vector embeddings generated by the `all-MiniLM-L6-v2` Sentence Transformer model. This enables the application to "understand" that a query for "remote roles" is contextually related to posts mentioning "work from home," even if the exact words do not overlap.

Key Features

- **Semantic Search Engine:** Uses neural embeddings to provide context-aware results.
- **Multi-Platform Integration:** Aggregates and filters posts from three major social networks.
- **Real-Time Analytics:** Displays search execution time and similarity scores for every result.
- **Dynamic Visual Identity:** Automatically generates author avatars based on initials.
- **Responsive Dashboard:** A modern, mobile-friendly interface built with Bootstrap 5.

Target Users

- **Social Media Managers:** To track industry trends and brand sentiment.
- **Data Researchers:** To extract qualitative insights from large social datasets.
- **Developers:** To integrate intelligent search capabilities into third-party apps via REST APIs.

Note: SocialMind AI is currently in its production-ready phase, optimized for high-performance retrieval across 300+ indexed records.

2. System Requirements

To ensure optimal performance of the Sentence Transformer models and the Flask web server, the following specifications are recommended.

Category	Minimum Requirement	Recommended
Processor	Dual-core CPU (2.0 GHz)	Quad-core CPU (2.5 GHz+)
RAM	4 GB	8 GB or higher
Storage	500 MB Free Space	1 GB Free Space
OS	Windows 10, macOS, Linux	Ubuntu 22.04 LTS or macOS Sonoma
Python	Version 3.9+	Version 3.10+
Browser	Chrome 90+, Firefox 88+	Chrome (Latest)

Tip: An active internet connection is required for the initial model download (approx. 80MB) and for rendering dynamic UI Avatars.

3. Installation Guide

Follow these steps to set up the development environment on your local machine.

Step 1: Clone the Repository

Open your terminal and clone the source code from GitHub.git clone https://github.com/VihitRaval/SocialMind-AI.git

cd SocialMind-AI

Step 2: Create a Virtual Environment

Initialize a clean environment to isolate project dependencies.python -m venv venv

For Windows

source venv/Scripts/activate

For macOS/Linux

source venv/bin/activate

Step 3: Install Dependencies

Install the required Python libraries, including Flask and Sentence Transformers.pip install -r requirements.txt

Warning: The `sentence-transformers` library may take several minutes to install as it downloads necessary PyTorch components.

4. Project Folder Structure

The application follows a modular architecture to separate business logic from the AI engine.

Directory/File	Purpose
`app.py`	Main entry point; handles Flask routing and server orchestration.
`models/`	Contains `semantic_search.py`, which manages the AI model and vectorization.
`database/`	Stores the `social_mind.db` SQLite file.

Directory/File	Purpose
`services/`	Orchestration layer that decouples search logic from the controllers.
`static/`	Holds frontend assets (CSS, JavaScript, and branding images).
`templates/`	Jinja2 HTML templates for the search interface and results rendering.
`data/`	Original JSON mock datasets used for database ingestion.

5. Running the Application Locally

Once the environment is configured, you can launch the platform.

1. **Activate Environment:** Ensure your virtual environment is active.
2. **Start Flask:** Execute the application script.python app.py
3. **Access Interface:** Open your web browser and navigate to: `http://127.0.0.1:5000`

6. Using the Application

6.1 Navigating the Homepage

The homepage features a centralized search bar with a "Hero-first" design. You will find navigation links to the Search, About, and Health monitoring pages in the top header.

[Placeholder: Screenshot of the Professional SaaS-style Homepage]

6.2 Executing a Search

Enter a natural language query in the search box (e.g., "AI career opportunities") and press enter.

- **Query Input:** The system accepts full sentences or phrases.
- **Validation:** If the input is empty, a JavaScript alert will prompt for a query.

6.3 Interpreting Results

Results are presented as interactive cards containing:

- **Similarity Score:** A percentage representing how closely the post matches your intent.
- **Dynamic Avatars:** Color-coded profiles based on the author's initials.
- **Platform Badge:** Indicates if the post is from LinkedIn, Facebook, or Instagram.

[Placeholder: Screenshot of Search Results Page with Similarity Scores and Cards]

7. REST API Guide

Developers can interact with the system programmatically through several RESTful endpoints.

`GET /`

- **Purpose:** Serves the main landing page.
- **Method:** `GET`
- **Response:** HTML Document.

`POST /search`

- **Purpose:** Process a semantic query and return ranked results.
- **Method:** `POST`
- **Parameters:** `query` (string).
- **Use Case:** Integrating the search engine into a mobile app or external dashboard.

`GET /health`

- **Purpose:** Monitors system status and AI model readiness.
- **Method:** `GET`
- **Response:** JSON object including model status and platform statistics.

8. Deployment Information

SocialMind AI is optimized for cloud environments using professional infrastructure.

- **Platform:** Render Cloud Platform.
- **Web Server:** Gunicorn (Green Unicorn) WSGI server for production stability.
- **CD Pipeline:** Every push to the GitHub repository triggers an automated build and deploy cycle on Render.
- **Live URL:** https://socialmind-ai-nmwg.onrender.com

9. Troubleshooting Guide

Problem	Possible Cause	Solution
App won't start	Python version mismatch	Ensure Python 3.9+ is installed.
ModuleNotFoundError	Dependencies not installed	Run `pip install -r requirements.txt` again.
Zero Results	Similarity threshold too high	Check query for extreme noise or random characters.
Images not loading	Internet connection issue	Ensure connectivity to `ui-avatars.com`.

Problem	Possible Cause	Solution
Port 5000 in use	Another Flask app is running	Kill the previous process or use `flask run --port 5001`.

10. Frequently Asked Questions (FAQ)

1. **What is semantic search?** It finds results based on meaning rather than exact keywords.
2. **Does it search live social media?** Currently, it uses a mock dataset of 300+ records for speed and stability.
3. **Is the code open source?** Yes, it is hosted on GitHub for community contribution.
4. **How are similarity scores calculated?** Using Cosine Similarity between 384-dimensional vectors.
5. **What model is used?** The `all-MiniLM-L6-v2` transformer model.
6. **Can I add more posts?** Yes, via the database ingestion scripts in the `data/` directory.
7. **Is it mobile responsive?** Yes, it uses Bootstrap 5 for a seamless mobile experience.
8. **What is the `/health` endpoint?** It provides real-time diagnostic data for the system.
9. **Why use SQLite?** It is lightweight and perfect for local prototyping and medium datasets.
10. **How long does a search take?** Typically under 0.5 seconds on production hardware.
11. **Do I need a GPU?** No, the model is optimized for CPU inference.
12. **What platforms are supported?** Instagram, Facebook, and LinkedIn.
13. **Can I use a different model?** Yes, the OOD architecture allows for model swapping in `semantic_search.py`.
14. **Is user data safe?** The system uses parameterized queries to prevent SQL injection.

15. **What happens if I search an empty string?** The system returns a validation error.
16. **How do avatars work?** They are generated dynamically via the UI Avatars API based on the author's name.
17. **Can I deploy this on AWS?** Yes, the structure is compatible with most cloud providers.
18. **What is Gunicorn?** A production-grade web server used for deployment on Render.
19. **How do I update dependencies?** Update `requirements.txt` and run the install command.
20. **Is there a dark mode?** Not currently, but it is planned for a future release.

11. Best Practices

- **Database Management:** Regularly back up the `social_mind.db` file before batch ingestion.
- **Dependency Updates:** Periodically check for updates to the `transformers` library for performance gains.
- **Querying:** For best results, use descriptive phrases instead of single keywords.
- **Deployment:** Always test locally before pushing changes to the `main` branch on GitHub.

12. Future Enhancements

The SocialMind AI roadmap includes several enterprise-grade features:

- **Live Integration:** Implementing official APIs for real-time post scraping.
- **Vector Databases:** Transitioning to Pinecone or Weaviate for million-record scale.
- **User Accounts:** Enabling "Favorite" searches and personalized search history.
- **Advanced Analytics:** Adding sentiment analysis and AI-generated post summarization.

13. Glossary of Technical Terms

- **Cosine Similarity:** A mathematical metric measuring the similarity between two vectors.

- **Embedding:** A numerical representation of text where similar meanings are close together.
- **Sentence Transformer:** A deep learning model specialized in converting text to vectors.
- **WSGI:** Web Server Gateway Interface, a standard for Python web application deployment.

14. Appendix

- **GitHub Repository:** https://github.com/VihitRaval/SocialMind-AI
- **Live Website:** https://socialmind-ai-nmwg.onrender.com
- **Primary Libraries:** Flask, Sentence Transformers, Scikit-learn, SQLite3.

15. Conclusion

SocialMind AI represents a significant advancement in social media information retrieval. By combining professional web engineering with state-of-the-art Natural Language Processing, the platform delivers a stable, scalable, and intuitive search experience. Whether used for academic evaluation or industrial research, it stands as a comprehensive demonstration of modern AI integration.

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